Module 6

1. What are the pillars of Wi-Fi security?

- **Authentication**
  Verifies the identity of users/devices before allowing access
- **Confidentiality**
  Protects data using encryption so others cannot read it
- **Integrity**
  Ensures data is not altered during transmission

2. Explain the difference between authentication and encryption in Wi-Fi security.

Authentication	Encryption
Verifies who the user/device is	Protects data from being read
Access control (allow/deny connection)	Data protection during transmission
Before connection is established	After connection (during data transfer)
Identity verification	Data confidentiality
Password login, 802.1X, SAE	AES
User is allowed or rejected	Data becomes unreadable to others

3. Explain the difference between WEP, WPA, WPA2 and WPA3?

Feature	WEP	WPA	WPA2	WPA3
Year	1997	2003	2004	2018
Encryption	RC4 (weak)	TKIP (RC4 improved)	AES-CCMP	AES / GCMP
Security Level	Very weak	Better than WEP	Strong	Strongest
Authentication	Open / Shared	PSK / 802.1X	PSK / 802.1X	SAE / 802.1X
Key Feature	Basic encryption	Dynamic keys	Strong encryption	Forward secrecy, better protection

4. Why is WEP considered insecure compared to WPA2 and WPA3?

- Uses weak RC4 encryption
- Uses small IV (24-bit) → repeats easily
- Static keys (do not change)
- No strong authentication
- Easy to crack using tools

5. Why was WPA2 introduced?

- WEP was very weak and easily cracked
- WPA improved security but still used TKIP (RC4-based) , not fully secure
- Needed a strong and long-term security solution
- WPA2 introduced AES-CCMP encryption (very strong)
- Provides better confidentiality, integrity, and authentication
- Supports both Personal (PSK) and Enterprise (802.1X) modes
- Became mandatory standard for Wi-Fi devices (from 2006)

6. What is the role of pairwise master key (PMK) in 4-way handshake?

- PMK (Pairwise Master Key) is the base key used in the handshake
- It is derived from:
    - PSK (Personal) or
    - EAP authentication (Enterprise)

- During 4-way handshake, PMK is used to:
    - Generate PTK (session key)
    - Verify both client and AP have the same key
    - Enable secure key exchange

7. How does the 4-way handshake ensure mutual authentication between the client and the access point?

- Both client and AP already have the PMK (shared secret)
- They exchange ANonce (AP) and SNonce (client)
- Both independently generate the same PTK
- Client sends MIC (Message Integrity Check) → AP verifies
- AP also sends MIC → client verifies

8. What Happens if a Wrong Passphrase is Entered During the 4-Way Handshake?

- Client and AP derive different PMK → different PTK
- Handshake starts normally (Message 1 & 2)
- MIC verification fails (keys don't match)
- Handshake stops → Message 3 & 4 not completed
- Connection is rejected / fails

9. What problem does 802.1X solve in the network?

- Eliminates shared password (PSK) problem
- Provides individual user authentication
- Uses centralized authentication (RADIUS server)
- Improves security and access control
- Prevents unauthorized users from accessing network
- Scales well for large networks (enterprise)

10. How 802.1X Enhances Security over Wireless Networks?

- Provides individual user authentication (no shared password)
- Uses RADIUS server (centralized control)
- Supports secure EAP methods
- Generates dynamic session keys (PMK, PTK)
- Enables mutual authentication (client + server verify each other)
- Allows access control (authorization) and user tracking
- Works with strong encryption (WPA2/WPA3)